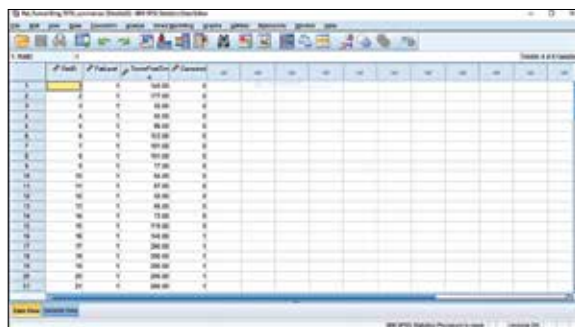


is a term used in statistics to indicate if a value is partially known. There are sub-classifications on censoring but we won't be getting into that level of detail. All that we need to know is that if the value in the censored column is 0 then it was uncensored data while a value of 1 indicates censored data. The experiment had 90 rats which were, in three batches of 30, fed three different diets with varying levels of fats. Fat level 1 indicates that the diet was low fat, level 2 indicates saturated fat diet and fat level 3 indicates an unsaturated fat diet.

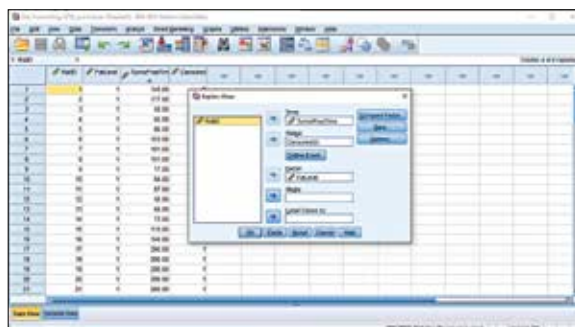
Fire up SPSS and open the provided data set. If you wish to understand how Kaplan-Meier estimator is applied to data sets then this video <http://dgjit.in/KaplanMeier> should help you out. Once you have the data loaded up, click on Analyze > Survival > Kaplan-Meier and a dialog box should pop up asking for the parameters. Here, select 'TumorFreeTime' as input for the 'Time'. Choose 'FatLevel' as input for the 'Factor' and then select the 'Censored' variable as input for the 'Status'. You will need to define the event for 'Status' so click on 'Define Event'. You now have three ways to define the 'Censored' value. As pointed out earlier, is the Censor value is 0 then the data is complete and the event has taken place. So we will be selecting 'Single value' as the manner in which we define the event and input '0' as the value. Click on Continue.

Now you need to compare the factor which is, in this case, the `FatLevel` variable. You are presented three different test statistics to compare the factor, we will be selecting Log rank, Breslow and Tarone-Ware. Click on Continue. Now click on Options to select which statistics and graph plots should be generated. Select Survival tables, Mean and Median survival, Survival plot and Hazard plots and then click Continue. Click OK to generate your report.

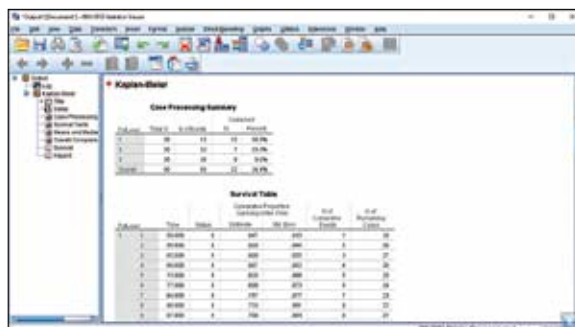
You are first presented with the Case Processing Summary which tells us that the rats that were on the low fat diet survived the longest and had the lowest cases of cancer developing. Fifteen rats did not develop tumours on FatLevel 1 diet, seven rats did not develop tumours on FatLevel 2 diet and all rats on FatLevel 3 diet ended up with tumours. As you scroll down,



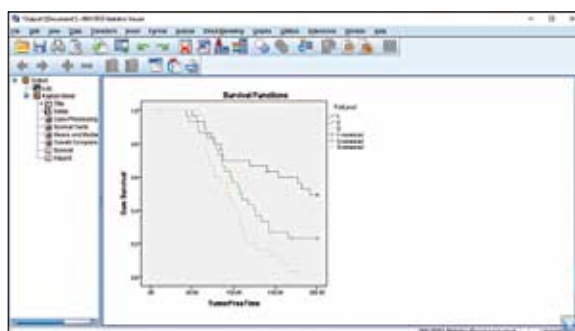
IBM SPSS with a data set loaded



### Applying Kaplan-Meier method



Your data has been analysed.



## Oooh! Graphs

you will come across the Survival Table which indicates at which point in time that the subjects developed tumour. The Means and Medians table shows the average time before rats on a certain diet would develop tumour.

And if tables aren't your thing then we've the same data in the form of easy to understand graphs. Each coloured line here is a cohort, or data pertaining

to a group of test subjects. The blue line shows rats on FatLevel 1 diet, green shows FatLevel 2 and the yellow line shows the unfortunate souls on FatLevel 3 diet. Based on the test data, you can conclude that unsaturated fat isn't good for rats.

You should know that we've used structured data for the above example and you're unlikely to get such structured data in real life situations, hence, the need for experienced data scientists to design data models for Big Data analysis.

## SOMETHING A LITTLE LESS COMPLICATED

If running your own experiments with SPSS or any other Big Data analytics tool isn't something you signed up for then you'd be surprised to know that you've probably been using or have reaped the benefits of services that use Big Data analytics for a few years now. If you've written any online article then you'd surely have performed a little Search Engine Optimisation and looked for keywords for your article. Well, you used Google to sift through millions of web pages to find keywords most appropriate for your article and that's Big Data analytics in action.

Have you seen those little factoids that pop onto the TV screen while enjoying your favourite sports match? How do you think those statistics are generated? Someone has to manually feed a key moment into the analytics system and the system will return those little facts. Now go on, and get cracking! 